

Jobs vs. Wages: A Data Visualization and Analytics Study of Unemployment and Median Household Income in the United States

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Many people assume that a lower unemployment rate directly translates into higher household incomes, since more people working should mean more money flowing into households. This assumption plays a big role in economic policy discussions, but it's rarely put to the test across all 50 U.S. states, Washington D.C., and Puerto Rico using one consistent dataset and method.

Does a lower unemployment rate truly predict a higher median household income across those locations? This question matters beyond academic interest, since where a person chooses to live and work often depends on how many jobs are available and how much they pay, particularly for first-generation students entering the workforce.

We hypothesized that unemployment rate would show a moderate to strong negative correlation with median household income, and that this single relationship would meaningfully predict income differences from state to state across the United States.

The primary objective was to test that hypothesis using real, population-weighted state-level economic data. If that relationship turned out to be weaker than expected, the goal was to find out which other commonly used economic factors do a better job explaining income differences across the country.

Data was drawn from the SimpleMaps uszips dataset (2022), which provides ZIP-code-level data on population, household income, unemployment rate, poverty rate and other demographic measures for the United States. Records were aggregated using population-weighted averages into 3,203 county-level rows and rolled up into 52 state and territory-level summaries. Pearson correlation coefficients and R-squared values were calculated between unemployment rate and median household income, with a linear trendline fitted to the data. The analysis was then expanded to all four attributes, unemployment rate, median household income, poverty rate, and college education rate, producing scatter plots with trendlines for each attribute pair and a full correlation matrix. Interactive visualizations including scatter plots, dynamic bar charts, and a comparison tool allowing up to five states or territories to be viewed simultaneously with individual trendlines were built using PHP, MySQL, and Google Charts, deployed to a live Kean University web server.

Contrary to the initial hypothesis, unemployment rate showed only a weak negative correlation with median household income ($r = -0.3753$, $R\text{-squared} = 0.1409$, $n = 52$), meaning unemployment explains just 14 percent of the variation in income across states and territories. Expanding the analysis revealed far stronger relationships, with unemployment correlating strongly with poverty rate ($r = 0.8532$), income correlating strongly with college education rate ($r = 0.7872$), and income correlating strongly and negatively with poverty rate ($r = -0.7366$). Puerto Rico emerged as the most extreme case, showing the highest unemployment and poverty rates alongside the lowest median household income of any state or territory in the dataset.

Unemployment rate alone is a weak predictor of household income, while poverty rate and college education rate are much stronger drivers of income differences across U.S. states and these territories. Two states can report identical unemployment rates while landing in very different income brackets, a gap best explained by poverty and education levels rather than employment status alone. These results suggest that policy efforts meant to improve household income may be more effective when they target poverty reduction and education access directly, rather than focusing on unemployment reduction in isolation.